

# GWAS Collaboration

## End-to-End Workflow Timeline

*Initiator • Collaboration Server • Collaborator*

Collaborative Study Web Application  
*Project documentation — prepared for review*

## How to read this document

Each section below is a single point in time (T0, T1, T2, ...) in the end-to-end GWAS collaboration workflow — from the very first account registration through the final AI-generated summary report.

Every time point shows three columns in parallel: what the Initiator is doing, what the Collaboration Server is doing, and what the Collaborator is doing. This makes it easy to see who is active when, and who is just waiting.

Cells that show a purple ► RUNS ON LOCAL MACHINE banner are doing real computation on that party's own computer. Raw genotype data NEVER leaves the local machine — only the aggregated survivor lists and case/control counts are returned to the Collaboration Server.

Below each swimlane is a set of screenshot placeholders — usually 3 to 5 per phase.

### Legend

|  |                       |                                                                                                 |
|--|-----------------------|-------------------------------------------------------------------------------------------------|
|  | Initiator             | The researcher who creates the collaboration, invites partners, and triggers GWAS + AI report.  |
|  | Collaboration Server  | Backend (Flask + MongoDB) handling auth, datasets, QC orchestration, chi-square, and LLM calls. |
|  | Collaborator          | An invited researcher who contributes their cohort + summary stats.                             |
|  | Screenshot box        | A dashed box showing the actual UI.                                                             |
|  | ► Runs locally banner | Indicates this party's work happens on their own machine. Raw data never leaves.                |

### Timeline at a glance

| Time | Phase                                      | What happens                         |
|------|--------------------------------------------|--------------------------------------|
| T0   | Account Registration                       | First-time setup                     |
| T1   | Login & Authentication                     | Returning session                    |
| T2   | Metadata / Dataset Upload                  | Each party registers their cohort    |
| T3   | Open Start Collaboration (Step 1: Details) | Initiator launches the 3-step wizard |
| T4   | Wizard Step 2 — Quality Control (with      | Choose QC methods + dataset          |

| Time       | Phase                                                   | What happens                                                         |
|------------|---------------------------------------------------------|----------------------------------------------------------------------|
|            | thresholds)                                             |                                                                      |
| <b>T5</b>  | <b>Wizard Step 3 — Find Collaborators &amp; Submit</b>  | <i>Search by phenotype + sample, invite, submit the wizard</i>       |
| <b>T6</b>  | <b>Review &amp; Accept Invitation</b>                   | <i>Collaborator agrees to participate</i>                            |
| <b>T7</b>  | <b>Automatic Chained Quality Control</b>                | <i>Server orchestrates — QC runs locally on each party's machine</i> |
| <b>T8</b>  | <b>Initiate QC Calculation, Threshold &amp; Preview</b> | <i>Pairwise QC: Initiate, live slider, confirm, preview list</i>     |
| <b>T9</b>  | <b>Build &amp; Upload GWAS Summary Stats</b>            | <i>Each party generates aggregated counts LOCALLY</i>                |
| <b>T10</b> | <b>Run GWAS Chi-Square</b>                              | <i>Initiator triggers aggregation</i>                                |
| <b>T11</b> | <b>View GWAS Results</b>                                | <i>Per-SNP statistics</i>                                            |
| <b>T12</b> | <b>Generate AI Summary Report</b>                       | <i>LLM-written, privacy-preserving</i>                               |
| <b>T13</b> | <b>Read &amp; Share AI Report</b>                       | <i>End of the workflow</i>                                           |

## T0 • Account Registration

*First-time setup*

| TIME                                 | INITIATOR                                                                                                                                                                                                                                                                                                                                               | COLLABORATION SERVER                                                                                                                                                                                                                                              | COLLABORATOR                                                                                                                                                                                               |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>T0</b><br><i>First-time setup</i> | <ul style="list-style-type: none"> <li>• Opens the web app at the public URL.</li> <li>• Clicks Register New Account on the login screen.</li> <li>• Fills in name, email, password, institution, phenotype interests.</li> <li>• Submits the form (POST /api/register).</li> <li>• Receives a confirmation and a JWT token in localStorage.</li> </ul> | <ul style="list-style-type: none"> <li>• Receives request.</li> <li>• Validates email uniqueness and password strength.</li> <li>• Hashes password and creates a record in the users collection.</li> <li>• Issues a JWT and returns it to the client.</li> </ul> | <ul style="list-style-type: none"> <li>• Independently visits the web app.</li> <li>• Clicks Register New Account and fills the same form.</li> <li>• Receives a JWT and is redirected to Home.</li> </ul> |

### Login / Register screen

*Welcome back, Login + Register New Account button.*

 Collaborative Study Web Application

[HOME](#) [LOGIN](#) [REGISTER](#)

Welcome back, Login

Email \*

Password \*



LOGIN

or

[FORGOT PASSWORD?](#)

Don't have an account?

[REGISTER NEW ACCOUNT](#)

## Registration form (Initiator)

*name, email, password.*

 Collaborative Study Web Application

[HOME](#) [LOGIN](#) [REGISTER](#)

### Registration

Name \*

Email \*

Password \*



Confirm Password \*

SUBMIT

Already have an account?

[LOGIN HERE](#)

*ste).*

## Registration form (Collaborator)

Same form, captured on the collaborator's machine to show both accounts.

 Collaborative Study Web Application

HOME LOGIN REGISTER

### Registration



SUBMIT

Already have an account?

LOGIN HERE

## T1 • Login & Authentication

Returning session

| TIME                                  | INITIATOR                                                                                                                                                                                                                                        | COLLABORATION SERVER                                                                                                                                                                                                                   | COLLABORATOR                                                                                                                               |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>T1</b><br><i>Returning session</i> | <ul style="list-style-type: none"> <li>Enter email + password on the login screen.</li> <li>Token is stored client-side; UI switches to the LoggedIn navbar.</li> <li>Lands on Home with quick links: Upload, Search, Collaborations.</li> </ul> | <ul style="list-style-type: none"> <li>validates credentials against the users collection.</li> <li>Returns { message: 'Login successful', token: &lt;JWT&gt; }.</li> <li>All subsequent calls are authenticated via token.</li> </ul> | <ul style="list-style-type: none"> <li>Logs in the same way from their own institution.</li> <li>Sees their own Home dashboard.</li> </ul> |

### Login screen (filled in)

Email + password entered, just before clicking Login.

 Collaborative Study Web Application
 HOME LOGIN REGISTER

### Welcome back, Login

Email \*

Password \*


or

[FORGOT PASSWORD?](#)

Don't have an account?

Logged-in Home dashboard (Initiator)

Collaborative Study Web Application

HOME METADATA COLLABORATION C

Welcome Back, Carter

Collaborations

PENDING (0) ACCEPTED (6) SENT (0)

No pending invitations.

Logged-in Home dashboard

Collaborative Study Web Application

HOME METADATA COLLABORATION S

Welcome Back, Sahith

Collaborations

PENDING (0) ACCEPTED (6) SENT (0)

Same view captured on the collaborator's machine — confirms both are signed in.

T2 • Metadata / Dataset

Each party registers their cohort

| TIME                                                       | INITIATOR                                                                                                                                                                                                                                                       | COLLABORATOR                                                                                                                                                              |
|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>T2</div> <div>Each party registers their cohort</div> | <div>► RUNS ON INITIATOR'S LOCAL MACHINE</div> <ul style="list-style-type: none"><li>Clicks MetaData home page)</li><li>Enters phenotype number of samples</li><li>Submits with multiple cohorts</li><li>Sees 'Metadata uploaded successfully' alert.</li></ul> | <div>IS ON COLLABORATOR'S LOCAL MACHINE</div> <p>forms the same upload from their own login.<br/>Their dataset metadata is now discoverable under the same phenotype.</p> |

## Metadata upload form (Initiator)

*Phenotype, # of Samples fields and Submit button.*

 Collaborative Study Web Application

[HOME](#) [METADATA](#) [COLLABORATION](#)



### Enter Metadata Details

Phenotype(s)

Heart Disease

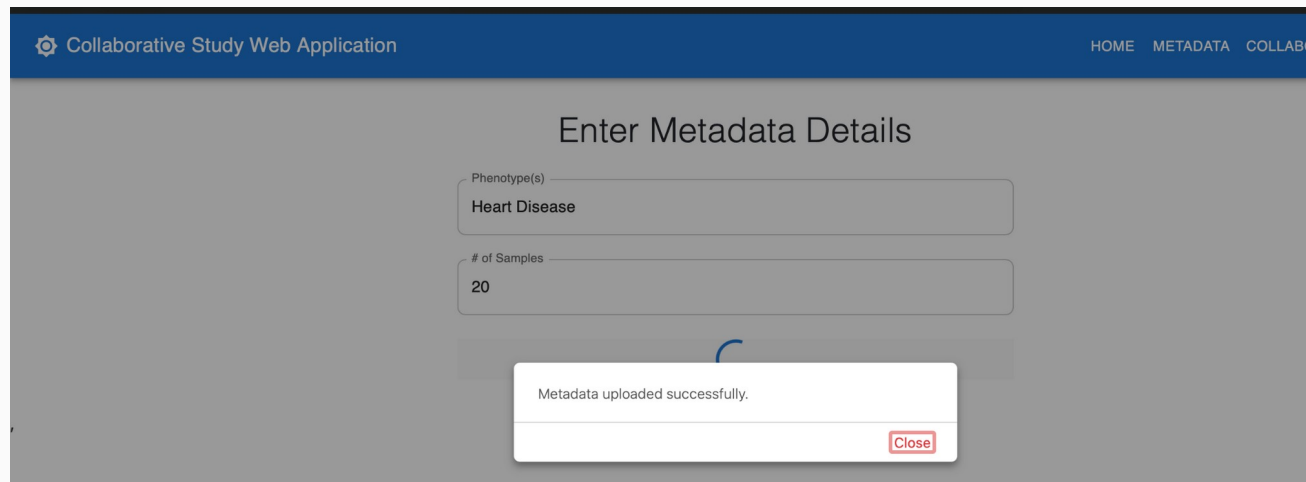
# of Samples

20

SUBMIT

## Upload success toast (Initiator)

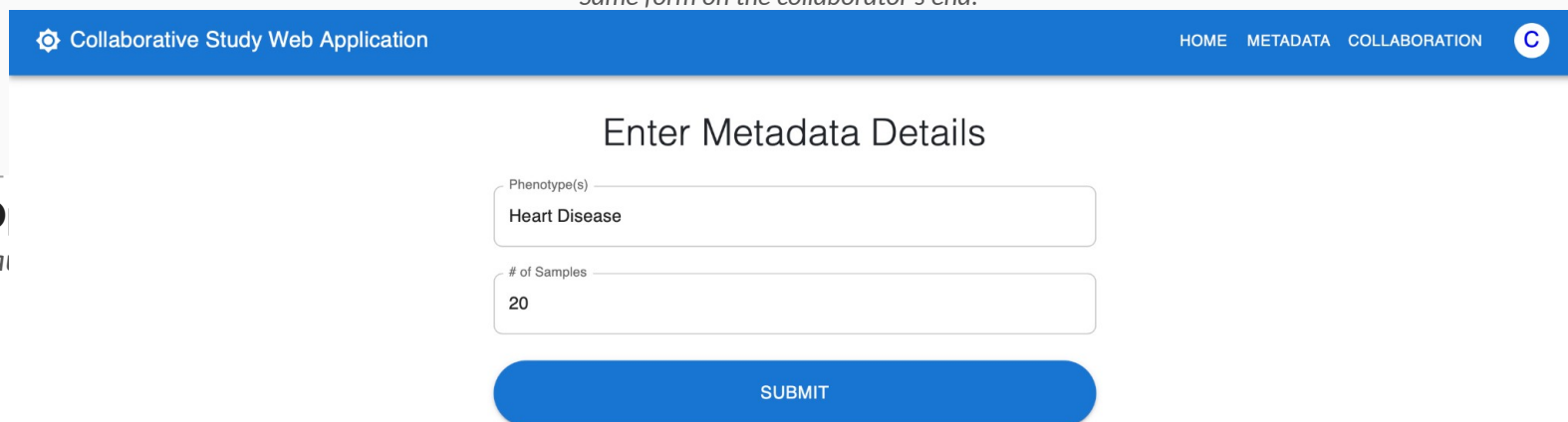
*alert/snackbar after a successful upload on the initiator's machine.*



The screenshot shows the 'Enter Metadata Details' form in the 'Collaborative Study Web Application'. The form has two input fields: 'Phenotype(s)' with the value 'Heart Disease' and '# of Samples' with the value '20'. A white toast message is displayed in the center, stating 'Metadata uploaded successfully.' with a 'Close' button in the bottom right corner. The application header includes 'HOME', 'METADATA', and 'COLLABORATION' links.

## Metadata upload form (Collaborator)

*Same form on the collaborator's end.*



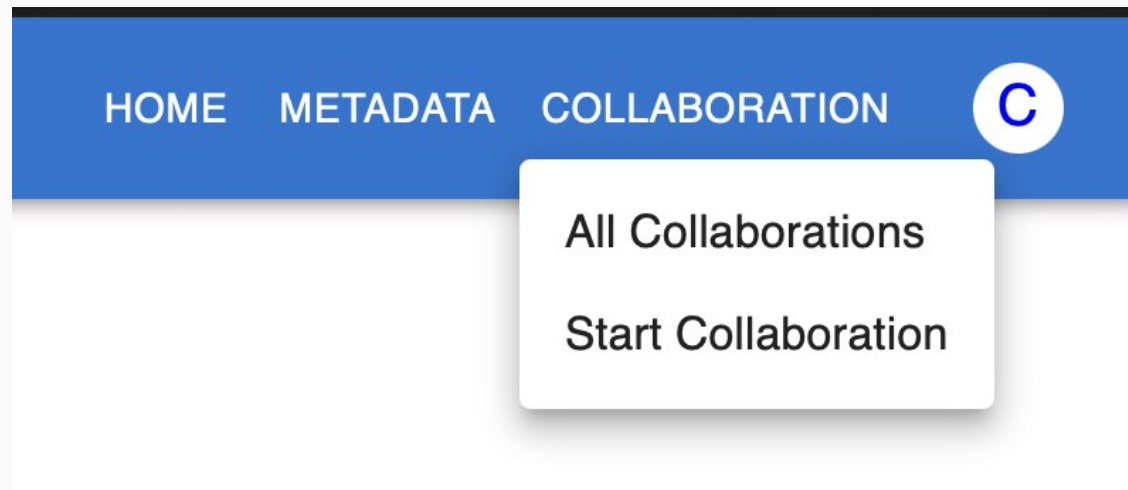
The screenshot shows the 'Enter Metadata Details' form in the 'Collaborative Study Web Application' for a collaborator. The form has two input fields: 'Phenotype(s)' with the value 'Heart Disease' and '# of Samples' with the value '20'. A large blue 'SUBMIT' button is at the bottom. The application header includes 'HOME', 'METADATA', 'COLLABORATION', and a user profile icon 'C'.

**T3 • O**  
*Initiator lai*

| TIME                                                     | INITIATOR                                                                                                                                                                                                                                                                                                                                             | COLLABORATION SERVER                                                                                                            | COLLABORATOR                                                                                                                            |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>T3</b><br><i>Initiator launches the 3-step wizard</i> | <ul style="list-style-type: none"> <li>Clicks Collaborations in the navbar to open the Collaborations page.</li> <li>Clicks Start a Collaboration to open the 3-step wizard.</li> <li>STEP 1 — Collaboration Details: types a Collaboration Name into the first field.</li> <li>Selects the Experiment Type from the dropdown (e.g. GWAS).</li> </ul> | <ul style="list-style-type: none"> <li>No backend call yet, The full submit happens only after Steps 2 + 3 (see T5).</li> </ul> | <ul style="list-style-type: none"> <li>Not active yet — has not been invited.</li> <li>Nothing to do on their side until T5.</li> </ul> |

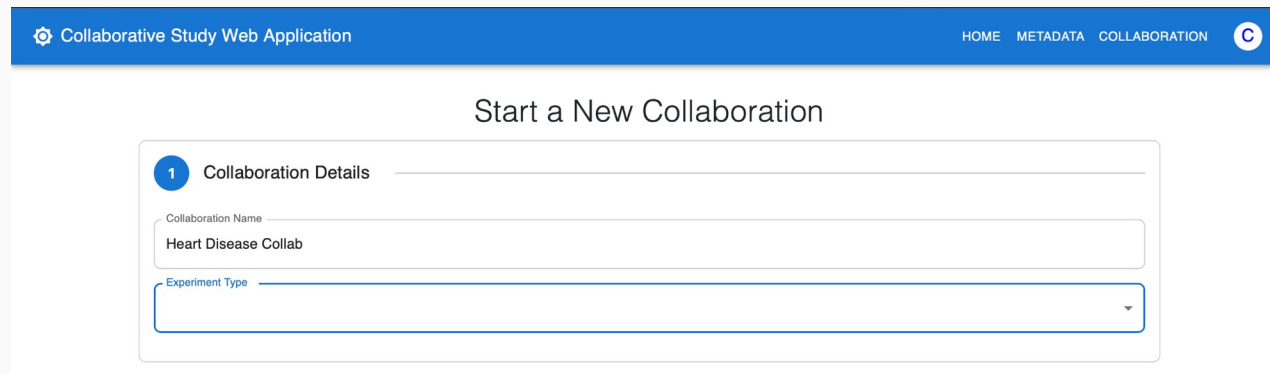
### Collaborations page (entry point)(Initiator)

*The Collaborations page just before clicking Start a Collaboration.*



## Wizard Step 1 — Collaboration Details (Name field)(Initiator)

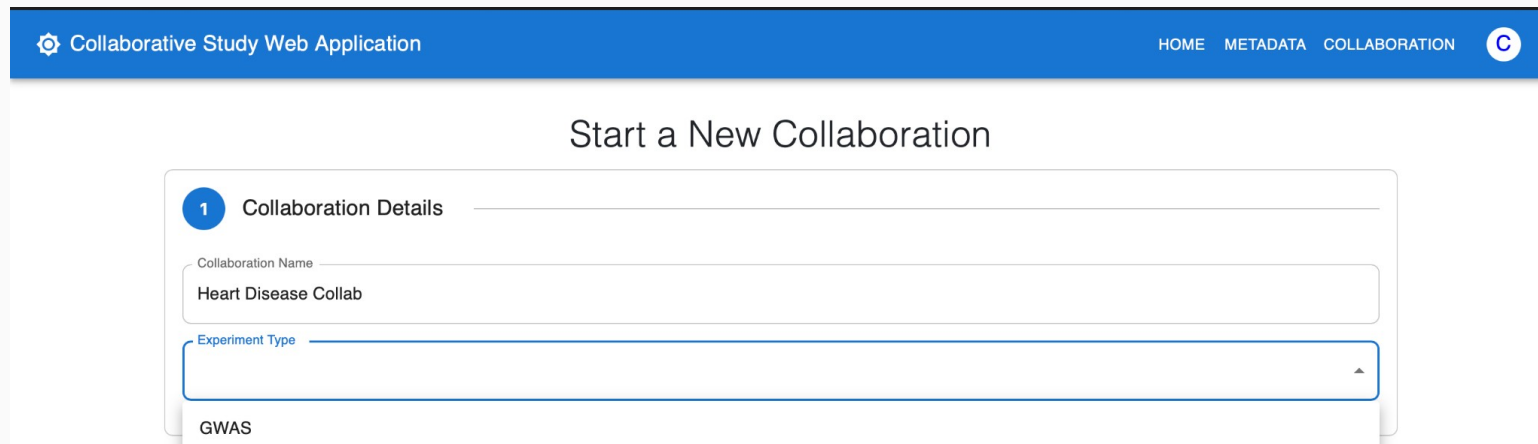
*Step 1: Collaboration Name input and Experiment Type closed dropdown.*



The screenshot shows the 'Start a New Collaboration' wizard. The top navigation bar is blue with a gear icon, the text 'Collaborative Study Web Application', and links for 'HOME', 'METADATA', 'COLLABORATION', and a user icon 'C'. The main heading is 'Start a New Collaboration'. Below it, a step indicator shows '1 Collaboration Details'. The form has two fields: 'Collaboration Name' with the value 'Heart Disease Collab' and 'Experiment Type' which is a closed dropdown menu.

## Experiment Type dropdown opened (Initiator)

*Experiment Type dropdown expanded showing GWAS option.*



The screenshot shows the 'Start a New Collaboration' wizard with the 'Experiment Type' dropdown menu open. The top navigation bar and heading are the same as the previous screenshot. The 'Collaboration Name' field still contains 'Heart Disease Collab'. The 'Experiment Type' dropdown is now expanded, showing a list of options with 'GWAS' visible at the bottom.

## T4 • Wizard Step 2 — Quality Control (with thresholds)

*Choose QC methods + dataset*

| TIME                                                       | INITIATOR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | COLLABORATION SERVER                                                                                                                                                                                           | COLLABORATOR                                                        |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <p><b>T4</b></p> <p><i>Choose QC methods + dataset</i></p> | <ul style="list-style-type: none"> <li>• STEP 2 — Quality Control: under “Choose QC Scheme (select multiple to chain)” picks one or more methods.</li> <li>• Available methods: Sample Relatedness, Population Stratification, Minor Allele Frequency (MAF), Hardy-Weinberg Equilibrium (HWE), Missing Data QC.</li> <li>• Multiple methods can be chained — they will run in sequence on each party's local machine (see T7).</li> <li>• Methods with thresholds (MAF cutoff, HWE p-value, missingness %) reveal a threshold input — initiator enters the cutoff values.</li> <li>• Picks an existing Dataset from the “Select Dataset” dropdown (the metadata uploaded at T2).</li> </ul> | <ul style="list-style-type: none"> <li>• Selections (methods + thresholds + chosen dataset) are buffered in the wizard state.</li> <li>• Stores the selected choices for the specific collaboration</li> </ul> | <ul style="list-style-type: none"> <li>• Not active yet.</li> </ul> |

## Wizard Step 2 — QC scheme cards (None selected) (Initiator)

Quality Control panel showing the row of selectable method chips with the default “None” selected.

Collaborative Study Web Application
HOME METADATA COLLABORATION C

### Start a New Collaboration

1
Collaboration Details

Collaboration Name  
Heart Disease Collab

Experiment Type

GWAS ✕

2
Quality Control

Choose QC Scheme (select multiple to chain)

☒ None
☐ Sample Relatedness
☐ Population Stratification
☐ Minor Allele Frequency (MAF)
☐ Hardy-Weinberg Equilibrium (HWE)

☐ Missing Data QC

Select Dataset

## Multiple QC methods selected to chain (Initiator)

*Several method chips selected together (e.g. MAF + Population Stratification) to show the chained-QC behaviour.*

Experiment Type

GWAS

2 Quality Control

Choose QC Scheme (select multiple to chain)

☐ None

☒ Sample Relatedness

☒ Population Stratification

☐ Minor Allele Frequency (MAF)

☐ Hardy-Weinberg Equilibrium (HWE)

☐ Missing Data QC

Select Dataset

## Threshold input for MAF / HWE / Missing-Data(Initiator)

Threshold fields that appear when MAF / HWE / Missing-Data are selected — capture with values typed in.

Experiment Type

GWAS ×

### 2 Quality Control

Choose QC Scheme (select multiple to chain)

☐ None

☐ Sample Relatedness

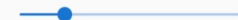
☒ Population Stratification

☒ Minor Allele Frequency (MAF)

☐ Hardy-Weinberg Equilibrium (HWE)

SNPs below this frequency are removed

MAF Threshold: 0.1



☐ Missing Data QC

Select Dataset

## Select Dataset dropdown( Initiator)

Select Dataset dropdown opened (or with a value chosen).

2

Quality Control

Choose QC Scheme (select multiple to chain)

☐ None
☐ Sample Relatedness
☒ Population Stratification
☒ Minor Allele Frequency (MAF)
☐ Hardy-Weinberg Equilibrium (HWE)

SNPs below this frequency are removed  
MAF Threshold: 0.1

☐ Missing Data QC

Select Dataset

Heart Disease | 20 samples

Bone Density | 156 samples

Bone Density | 157 samples

Heart Disease | 156 samples

Heart Disease | 111 samples

Heart Disease | 157 samples

Phenotype(s)

Minimum # of Samples

SEARCH

Discover collaborators to partner with on your new experiment.

## T5 • Wizard Step 3 — Find Collaborators & Submit

*Search by phenotype + sample, invite, submit the wizard*

| TIME                                                                        | INITIATOR                                                                                                                                                                                                                                                                                         | COLLABORATION SERVER                                                                                                                                                                                                                                                                     | COLLABORATOR                                                                                                                                                           |
|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>T5</b><br><i>Search by phenotype + sample, invite, submit the wizard</i> | <ul style="list-style-type: none"> <li>STEP 3 — Find Collaborators: searches for partners by Phenotype + Sample Size from within the wizard.</li> <li>Clicks Invite on each chosen researcher — they appear in the Invited list with status “pending”.</li> <li>Clicks Submit / Create</li> </ul> | <ul style="list-style-type: none"> <li>Receives start collaboration request; creates the collaboration document (status = pending, stores creator_id, name, experiment, QC schemes + thresholds, dataset, UUID).</li> <li>For each invitee, it sends the collaboration invite</li> </ul> | <ul style="list-style-type: none"> <li>A notification badge appears on their navbar.</li> <li>Sees the new pending invitation on their Collaborations page.</li> </ul> |

### Wizard Step 3 — Find Collaborators search inputs ( Initiator)

*Step 3 — Phenotype + Sample Size search inputs visible.*

3 Find Collaborators

Phenotype(s)  
heart Disease

Minimum # of Samples

SEARCH

## Search results inside the wizard ( Initiator)

Search result cards (researcher name + phenotype + sample count) inside the wizard.

### 3 Find Collaborators

Phenotype(s)

heart Disease

Minimum # of Samples

SEARCH

#### Heart Disease

Number of Samples: 100000  
By: Dr. Cheng Rui

☐ Select for Collaboration

#### Heart Disease

Number of Samples: 20000  
By: Dr. Evelyn Carter

☐ Select for Collaboration

#### PlaqueTangle Alzheimer Disease

Number of Samples: 13500  
By: Prof. Matthew Wilson

☐ Select for Collaboration

#### Alzheimer Disease - PHENOME

Number of Samples: 5300  
By: Dr. Lydia

☐ Select for Collaboration

#### Heart Disease

Number of Samples: 157  
By: John

☐ Select for Collaboration

#### Heart Disease

Number of Samples: 20  
By: Sahith

☒ Select for Collaboration

#### heart D

Number of Samples: 20  
By: Sahith

☐ Select for Collaboration

#### Heart Disease

Number of Samples: 20  
By: james

☐ Select for Collaboration

CREATE COLLABORATION

## T6 • Review & Accept Invitation

*Collaborator agrees to participate*

| TIME                                                   | INITIATOR                                                                                                                                                                                       | COLLABORATION SERVER                                                                                                                                                                                                                                           | COLLABORATOR                                                                                                                                                                                           |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>T6</b><br><i>Collaborator agrees to participate</i> | <ul style="list-style-type: none"> <li>Refreshes the collaboration detail page and watches the invited list.</li> <li>Sees the collaborator's status change from pending → accepted.</li> </ul> | <ul style="list-style-type: none"> <li>Receives request.</li> <li>Updates invited_users status = 'accepted' on the collaboration document.</li> <li>When the last invitee has responded, internally calls methods to trigger qc data set generation</li> </ul> | <ul style="list-style-type: none"> <li>Opens Home Page for the pending invite.</li> <li>Reviews initiator name, QC scheme, phenotype, expected dataset.</li> <li>Clicks Accept (or Reject).</li> </ul> |

### Invited list with pending status ( Initiator)

*Invited list in the Homepage, with each invitee showing pending.*

Collaborative Study Web Application
HOME METADATA COLLABORATION C

Collaborations

PENDING (0)

ACCEPTED (6)

SENT (1)

Heart Disease Collab  
Initiated by You

QUICK VIEW
VIEW DETAILS

## Collaborator's side + pending invite card

Collaborator's machine: notification badge in the navbar + the new pending invite card on the Collaborations page.

 Collaborative Study Web Application

HOME METADATA COLLABORATION 

# Welcome Back, Sahith

## Collaborations

PENDING (1)

ACCEPTED (6)

SENT (0)

 **Heart Disease Collab**  
Initiated by Carter

[QUICK VIEW](#) [VIEW DETAILS](#)

Collaboration details — invite preview (Collaborator)

Quick View about the collaboration that contains all the details

 Collaborative Study Web Application

HOME METADATA COLLABORATION 

# Welcome Back, Sahith

## Collaborations

PENDING (1)

ACCEPTED (6)

SENT (0)

 **Heart Disease Collab**  
Initiated by Carter

QUICK VIEW

VIEW DETAILS

Participants

S Sahith

✓

ACCEPT

✕

REJECT

Experiments

GWAS

QC Schemes

Population Stratification

Minor Allele Frequency (MAF)

Your Dataset

Phenotype: Heart Disease

Samples: 20



## Status flips to accepted (Initiator's view)

*Initiator refreshes their detail page — the row shows 'accepted'.*

 Collaborative Study Web Application

HOME METADATA COLLABORATION 

### Collaboration Details

**Title:** Heart Disease Collab

**Experiments**

GWAS

**Phenotypes & Number of Samples**

Heart Disease - 20 Heart Disease - 20

**QC Scheme**

Population Stratification Minor Allele Frequency (MAF)

Collaborator: Sahith

**Collaborators**

Carter (Initiator)

Sahith (Collaborator) Accepted

**Collaboration Status**

✓ Onboarding Collaborators

2 QC Calculation

## T7 • Automatic Chained Quality Control

*Server orchestrates — QC runs locally on each party's machine*

| TIME                                                                                              | INITIATOR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | COLLABORATION SERVER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | COLLABORATOR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>T7</b><br><i>Server<br/>orchestrates —<br/>QC runs locally<br/>on each party's<br/>machine</i> | <p>► RUNS ON INITIATOR'S LOCAL MACHINE</p> <ul style="list-style-type: none"> <li>Initiator's local QC controller is running in the background on their machine.</li> <li>When the server pings it, the controller reads the initiator's RAW genotype data locally — raw data NEVER leaves this machine.</li> <li>Runs MAF → HWE → Missing-Data → Sample Relatedness → Population Stratification in sequence (based on what was selected).</li> <li>Returns ONLY surviving sample IDs and SNP IDs back to the orchestrator (no genotypes).</li> <li>On the browser UI, the initiator sees the QC Calculation panel update to 'In progress' then 'Done'.</li> <li>Then the initiator has an option to initiate QC calculation once the data set is ready (in case of population stratification and sample relatedness)</li> </ul> | <ul style="list-style-type: none"> <li>Orchestrator (Flask app) detects all invitees have accepted and triggers _qc data set creation methods(uuid).</li> <li>Filter-only methods (MAF / HWE / Missing-Data) are dispatched first, then pairwise methods (Relatedness / Population Stratification).</li> <li>For each step, the server sends request to each party's local QC controller at with { username, phenotype, method }.</li> <li>Polls each controller for completion; once done, writes surviving_samples and surviving_snps per user into the collaborations document.</li> <li>The server never sees raw genotypes — only survivor lists.</li> </ul> | <p>► RUNS ON COLLABORATOR'S LOCAL MACHINE</p> <ul style="list-style-type: none"> <li>Collaborator's local QC controller is also running in the background on their machine.</li> <li>Receives the same QC requests from the orchestrator and runs them against the collaborator's OWN raw data — locally.</li> <li>Raw data never leaves the collaborator's machine; only survivor lists are returned.</li> <li>On the browser UI, the collaborator sees the same live QC panel.</li> </ul> |
|                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## Local QC controller terminal/logs (Initiator)

*Terminal window where qc\_controller.py is running on the initiator's machine — capture the log lines that print as each method runs.*

```
=====
📧 New QC Request:
Request ID: 69bc9624f0a13eb1e925c05d-1779150516424
User: Carter (69bc9624f0a13eb1e925c05d)
Action: create_chained
Phenotype: Heart Disease
Method: chained
=====

✅ [Orchestrator] Started QC container: qc-carter-1779150516428-nxvhl4gv for user Carter
Running: 1/10
=====
```

## Local QC controller terminal/logs (Collaborator)

*Same terminal window on the collaborator's machine.*

```
=====
📧 New QC Request:
Request ID: 682201f4c18f834bb93641a7-1779150516594
User: Sahith (682201f4c18f834bb93641a7)
Action: create_chained
Phenotype: Heart Disease
Method: chained
=====

✅ [Orchestrator] Started QC container: qc-sahith-1779150516594-bnt26fgg6 for user Sahith
Running: 2/10
=====
```

## QC Calculation panel — running/after running( initiator)

Browser UI: progress bar + per-method status chips (Initiator).

Collaborative Study Web Application
 

HOME METADATA COLLABORATION

C

### Collaboration Details

**Title:** Heart Disease Collab

**Experiments**

GWAS

**Phenotypes & Number of Samples**

Heart Disease - 20

Heart Disease - 20

**QC Scheme**

Population Stratification

Minor Allele Frequency (MAF)

**Quality Control**

INITIATE QC CALCULATION

**Collaborators**

Carter (Initiator)

Sahith (Collaborator) Accepted

**Collaboration Status**

✓ Onboarding Collaborators

2 QC Calculation

Awaiting the initiator to start the QC calculation

3 Stat Data

4 Final Experiment

## QC Calculation panel — running/after running (Collaborator)

Same browser panel viewed on the collaborator's machine.

 Collaborative Study Web Application

HOME METADATA COLLABORATION 

### Collaboration Details

**Title:** Heart Disease Collab

**Experiments**

GWAS

**Phenotypes & Number of Samples**

Heart Disease - 20 Heart Disease - 20

**QC Scheme**

Population Stratification Minor Allele Frequency (MAF)

**Collaborators**

Carter (Initiator)

Sahith (Collaborator) Accepted

**Collaboration Status**

✓ Onboarding Collaborators

2 QC Calculation  
Awaiting the initiator to start the QC calculation

3 Stat Data

4 Final Experiment

## T8 • Initiate QC Calculation, Threshold & Preview

Pairwise QC: Initiate, live slider, confirm, preview list

| TIME                                                                                           | INITIATOR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | COLLABORATION SERVER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | COLLABORATOR                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>T8</div> <div>Pairwise QC:<br/>Initiate, live<br/>slider, confirm,<br/>preview list</div> | <ul style="list-style-type: none"><li>• PAIRWISE QC ONLY (Sample Relatedness, Population Stratification): clicks the “Initiate QC Calculation” button on the QC Calculation panel (POST /api/calculate-qc). Until clicked, the UI shows “Awaiting the initiator to start the QC calculation” and NO pairwise counts are visible.</li><li>• A dynamic Threshold Slider appears (MUI Slider, range 0—1, step 0.01). As the initiator drags, per-party “samples to be included” counts update LIVE entirely client-side — a per-user histogram re-renders in real time. No API call per drag.</li><li>• Clicks Confirm Threshold (enabled only after the slider moves) — POST /api/datasets/&lt;uuid&gt;/qc-results { threshold, qc_scheme }. The per-party preview list of surviving sample IDs is then revealed and locked in for downstream GWAS.</li></ul> | <ul style="list-style-type: none"><li>• POST /api/calculate-qc runs Sample Relatedness (kinship phi_value via compute_coefficients_array) and Population Stratification (Euclidean PCA distance via calculate_pairwise_distances_pca) across all parties. Returns the full pairwise comparison set { user1, sample1, user2, sample2, phi_value   distance }.</li><li>• GET /api/datasets/&lt;uuid&gt;/qc-results streams those comparisons to the browser; the slider then filters them in-memory — no per-drag server call.</li><li>• On Confirm Threshold POST, the server stores threshold in the collaboration doc, re-filters with phi_value ≤ t or distance ≤ t, groups survivors per user into filtered_qc_sample_relatedness / filtered_qc_population_stratification, and returns the final per-party sample lists.</li></ul> | <ul style="list-style-type: none"><li>• Read-only — sees the same threshold slider and live per-party counts as the initiator. Only the initiator can click Confirm Threshold.</li><li>• Once the initiator confirms, the collaborator sees the same per-party preview sample list that will be used downstream by GWAS.</li></ul> |

[ SCREENSHOT PLACEHOLDER ]

### Initiate QC Calculation button (idle, before click)

CollaborationDetails.jsx — the “Initiate QC Calculation” button visible only when pairwise QC is selected and not yet run.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[ SCREENSHOT PLACEHOLDER ]

### Dynamic threshold slider with live per-party counts

*MUI Slider (range 0-1, step 0.01) with the per-user histogram of “samples to be included” updating as the slider moves.*

*Replace this box with your screenshot (right-click → Insert Picture, or paste).*

[ SCREENSHOT PLACEHOLDER ]

### Confirm Threshold + per-party preview sample list

*After clicking Confirm Threshold, the panel below shows each party's surviving sample IDs that will feed the GWAS step.*

*Replace this box with your screenshot (right-click → Insert Picture, or paste).*

## T9 • Build & Upload GWAS Summary Stats

Each party generates aggregated counts LOCALLY

| TIME                                                                    | INITIATOR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | COLLABORATION SERVER                                                                                                                                                                                                                                                                                                                                                                                            | COLLABORATOR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>T9</div> <div>Each party generates aggregated counts LOCALLY</div> | <div>▶ RUNS ON INITIATOR'S LOCAL MACHINE</div> <ul style="list-style-type: none"><li>Initiator clicks Create GWAS Dataset in the browser UI.</li><li>The browser POSTs /api/create_gwas_dataset to the server, which in turn asks the initiator's LOCAL controller to compute case/control counts per SNP using the QC-filtered sample list — this read happens on the initiator's machine, against the initiator's raw genotypes.</li><li>Only the aggregated counts (SNP_id → {case_count, control_count}) leave the local machine.</li><li>Alternatively, the initiator can manually upload a pre-computed stats CSV via POST /api/upload_csv_stats.</li><li>Upload flag flips to true on their row.</li></ul> | <ul style="list-style-type: none"><li>Coordinates the request — triggers each party's local controller to generate stats.</li><li>Receives aggregated case/control counts only (never raw genotypes).</li><li>Stores stats in the datasets collection (is_qc_data = False, with case/control counts per SNP).</li><li>When all expected users have uploaded, sets collaboration.stat_uploaded = true.</li></ul> | <div>▶ RUNS ON COLLABORATOR'S LOCAL MACHINE</div> <ul style="list-style-type: none"><li>Collaborator clicks the same Create GWAS Dataset button in their browser UI.</li><li>Their LOCAL controller computes case/control counts from their own QC-filtered samples — again, raw data stays local.</li><li>Only aggregated counts are returned to the server.</li><li>Or: collaborator uploads a stats CSV manually.</li><li>Their row in the upload-status table flips to 'uploaded'.</li></ul> |

[ SCREENSHOT PLACEHOLDER ]

### Stat Data Upload panel (Initiator)

CollaborationDetails.jsx — Create GWAS Dataset + Upload CSV options.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[ SCREENSHOT PLACEHOLDER ]

### Create GWAS Dataset — confirmation dialog

*The modal/loading state when the initiator triggers stat generation.*

*Replace this box with your screenshot (right-click → Insert Picture, or paste).*

[ SCREENSHOT PLACEHOLDER ]

### Local controller logs generating stats (Initiator)

*Terminal window on the initiator's machine showing case/control counts being computed.*

*Replace this box with your screenshot (right-click → Insert Picture, or paste).*

[ SCREENSHOT PLACEHOLDER ]

### Local controller logs generating stats (Collaborator)

*Same terminal capture on the collaborator's machine.*

*Replace this box with your screenshot (right-click → Insert Picture, or paste).*

[ SCREENSHOT PLACEHOLDER ]

### Stat upload status per party

*Table showing initiator + each collaborator with Uploaded / Pending badges.*

*Replace this box with your screenshot (right-click → Insert Picture, or paste).*

## T10 • Run GWAS Chi-Square

*Initiator triggers aggregation*

| TIME                                                | INITIATOR                                                                                                                                                                                                                                                              | COLLABORATION SERVER                                                                                                                                                                                                                                                                                | COLLABORATOR                                                                                                 |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>T10</b><br><i>Initiator triggers aggregation</i> | <ul style="list-style-type: none"> <li>Goes to the Experiment Results section.</li> <li>Clicks Initiate GWAS Calculation (enabled only once stat_uploaded = true).</li> <li>POST /api/calculate_chi_square { uuid }.</li> <li>UI shows 'computing' spinner.</li> </ul> | <ul style="list-style-type: none"> <li>Aggregates case/control counts across initiator + all accepted collaborators.</li> <li>Builds contingency tables per SNP and computes chi-square + p-value.</li> <li>Stores per-user and aggregated results in collaborations.chi_square_results.</li> </ul> | <ul style="list-style-type: none"> <li>Read-only — watches the same spinner / progress indicator.</li> </ul> |

[ SCREENSHOT PLACEHOLDER ]

### Experiment Results panel — pre-compute

*Initiate GWAS Calculation button.*

*Replace this box with your screenshot (right-click → Insert Picture, or paste).*

[ SCREENSHOT PLACEHOLDER ]

### Confirmation dialog

*Optional modal asking the initiator to confirm before computing.*

*Replace this box with your screenshot (right-click → Insert Picture, or paste).*

[ SCREENSHOT PLACEHOLDER ]

**Compute in progress**

*Spinner + status text while chi-square runs.*

*Replace this box with your screenshot (right-click → Insert Picture, or paste).*

## T11 • View GWAS Results

Per-SNP statistics

| TIME                                         | INITIATOR                                                                                                                                                                                                                                                        | COLLABORATION SERVER                                                                                                                                                                               | COLLABORATOR                                                                                                                                                                |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>T11</div> <div>Per-SNP statistics</div> | <ul style="list-style-type: none"><li>• GET /api/calculate_chi_square_results/&lt;uuid&gt; populates a results table.</li><li>• Sees SNPs ranked by p-value with chi-square, OR, and per-site contribution.</li><li>• Can download the results as CSV.</li></ul> | <ul style="list-style-type: none"><li>• Serves cached chi-square results from the collaboration document.</li><li>• Computes any derived stats requested by the UI (sorting, filtering).</li></ul> | <ul style="list-style-type: none"><li>• Sees the same results table from their CollaborationDetails view.</li><li>• Has the same export options as the initiator.</li></ul> |

[ SCREENSHOT PLACEHOLDER ]

GWAS results table — top SNPs (Initiator)

Sortable table of chi-square / p-value.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[ SCREENSHOT PLACEHOLDER ]

GWAS results table (Collaborator)

Same results table viewed on the collaborator's machine.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[ SCREENSHOT PLACEHOLDER ]

### Manhattan-style or bar chart of $-\log_{10}(p)$

*If your UI renders a chart for results, snapshot it here.*

*Replace this box with your screenshot (right-click → Insert Picture, or paste).*

[ SCREENSHOT PLACEHOLDER ]

### Export / Download CSV button

*Wherever the user can save the results locally.*

*Replace this box with your screenshot (right-click → Insert Picture, or paste).*

## T12 • Generate AI Summary Report

LLM-written, privacy-preserving

| TIME                                                          | INITIATOR                                                                                                                                                                                                                                               | COLLABORATION SERVER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | COLLABORATOR                                                                                                                                                              |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>T12</div> <div>LLM-written,<br/>privacy-preserving</div> | <ul style="list-style-type: none"><li>• Opens GwasSummaryCard inside CollaborationDetails.</li><li>• Clicks Generate Summary — POST /api/collaboration/&lt;uuid&gt;/gwas-summary.</li><li>• While it runs, the UI shows a 'generating' state.</li></ul> | <ul style="list-style-type: none"><li>• Builds a privacy-preserving digest from chi_square_results: anonymises collaborators as 'Site A', 'Site B', etc. — no raw genotypes, no participant IDs.</li><li>• Calls the LLM (gpt-4o-mini) with the digest and a structured system prompt.</li><li>• Parses the structured JSON: { overview, top_snps, per_collaborator, recommendation: continue   continue_with_caveats   reconsider }.</li><li>• Stores in collaborations.ai_summary with generated_at and site_label_map.</li></ul> | <ul style="list-style-type: none"><li>• Read-only — can view the summary once it's generated.</li><li>• Sees themselves only under their anonymised site label.</li></ul> |

[ SCREENSHOT PLACEHOLDER ]

Generate Summary button (idle)

GwasSummaryCard before clicking.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[ SCREENSHOT PLACEHOLDER ]

### Loading state

*Spinner / 'Generating...' state while the LLM runs.*

*Replace this box with your screenshot (right-click → Insert Picture, or paste).*

[ SCREENSHOT PLACEHOLDER ]

### Server log of outbound LLM call (optional)

*Flask app terminal showing the POST to OpenAI being made — proves only aggregated digest is sent.*

*Replace this box with your screenshot (right-click → Insert Picture, or paste).*

## T13 • Read & Share AI Report

End of the workflow

| TIME                                          | INITIATOR                                                                                                                                                                                                                                                                                                                                        | COLLABORATION SERVER                                                                                                                                                                                                       | COLLABORATOR                                                                                                                                                                                                                          |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>T13</div> <div>End of the workflow</div> | <ul style="list-style-type: none"><li>Reads the final report: overview paragraph, ranked top SNPs, per-site contribution, recommendation callout.</li><li>GET /api/collaboration/&lt;uuid&gt;/gwas-summary returns the cached summary on later visits.</li><li>Can copy / export the report and discuss next steps with collaborators.</li></ul> | <ul style="list-style-type: none"><li>Caches the report in collaborations.ai_summary; subsequent GETs are read-only.</li><li>Closes the workflow — no more compute is required unless the initiator regenerates.</li></ul> | <ul style="list-style-type: none"><li>Opens CollaborationDetails and sees the same final report.</li><li>Identified only by site label (privacy preserved).</li><li>Can use the recommendation to decide on follow-up work.</li></ul> |

[ SCREENSHOT PLACEHOLDER ]

Final AI Summary Card — top of report

Overview + Top SNPs section.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[ SCREENSHOT PLACEHOLDER ]

Final AI Summary Card — per-site analysis

The middle block of the summary card.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[ SCREENSHOT PLACEHOLDER ]

**Recommendation banner (continue / caveats / reconsider)**

*The colored callout at the bottom of the card.*

*Replace this box with your screenshot (right-click → Insert Picture, or paste).*

[ SCREENSHOT PLACEHOLDER ]

**Same report on Collaborator's screen**

*Confirms collaborators see only site labels, not real names.*

*Replace this box with your screenshot (right-click → Insert Picture, or paste).*

## How to capture local-machine screenshots

Some steps (T2, T7, T9) run on each party's own computer — the local QC controller at localhost:5001 does the actual data crunching. Here is how to capture evidence of that local activity so the placeholders in those sections can show real screenshots.

### 1. Terminal running `qc_controller.py`

On the machine that's playing the role (Initiator or Collaborator), start the controller with ``python qc_controller.py`` from `Collaborator_Server/`. Leave the Terminal window visible. Trigger the QC from the web UI and screenshot the Terminal as the log lines scroll by: 'Running MAF...', 'Surviving samples: N', etc. macOS: Cmd+Shift+4 + Space to capture just the window. Windows: Snip & Sketch (Win+Shift+S).

### 2. Browser DevTools → Network tab

Open the web app, hit F12 / Cmd+Option+I, switch to the Network tab, filter by `'/api/qc'` or `'/api/create_gwas_dataset'`. Trigger the action. Screenshot the network panel showing the request being POSTed to localhost:5001 — this proves the orchestrator is talking to the local controller, not pulling raw data into the cloud.

### 3. Raw data file in Finder / Explorer

Open the folder where the party's raw genotype CSV lives (e.g. `~/data/eye_color.csv`). Screenshot the Finder/Explorer window showing the file timestamp before and after the QC run is unchanged — emphasises that QC reads it but doesn't move it.

### 4. Flask app server log

On whichever machine the Collaboration Server is running, screenshot the Flask log lines showing the orchestrator firing requests to each party's QC controller URL. This is what proves the server is coordinating, not computing on raw data itself.

### 5. Side-by-side composite

For the most impressive figure, take two screenshots at the same moment — one of the Initiator's local Terminal and one of the Collaborator's local Terminal — and place them side by side in the doc. This visually demonstrates that both parties are crunching their own data in parallel.

## How to drop multiple screenshots into one section

Each phase already has several dashed placeholder boxes. Click inside a box, delete the placeholder text, and use Insert → Pictures → This Device. If you need more boxes, copy any existing dashed box (click its border, Cmd/Ctrl+C, Cmd/Ctrl+V) and paste it below — Word preserves the formatting.

## Notes on privacy and data handling

- Raw genotype data never leaves a party's local environment. Only QC survivor lists and per-site case/control counts are shared with the Collaboration Server.
- The chi-square aggregation uses contingency tables only — there is no row-level genotype exchange between Initiator and Collaborator.
- The AI summary is built from an already-aggregated digest. Collaborators are anonymised as Site A / Site B / ... in both the prompt and the rendered report; participant IDs are never sent to the LLM.
- The generated summary is cached server-side so that subsequent reads (GET /api/collaboration/<uuid>/gwas-summary) do not re-call the LLM.